

CyberShield AI

Solution Report

AI-Powered Behavioral Anomaly Detection for Cybersecurity

Honeywell Hackathon 2026

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Deployed Website: <https://cybershield-ai-1-5gyk.onrender.com/>

GitHub Repository: <https://github.com/Prasanth4689/CyberShield-AI>

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1. Executive Summary

CyberShield AI is an end-to-end behavioral anomaly detection system built for enterprise cybersecurity. It features a 4-layer machine learning pipeline profiling 200+ entities (users, service accounts, edge devices), detecting 7 attack types, and serving an analyst dashboard deployed live at <https://cybershield-ai-1-5gyk.onrender.com/>.

Performance metrics: Accuracy = 93.6%, Precision = 75.2%, Recall = 97.7%, F1 Score = 85.0%, AUC-ROC = 0.9878.

2. Problem Analysis

Static, rule-based SIEM systems fail in modern OT/IT environments like Honeywell's due to high false positive rates (up to 50%), inability to detect novel zero-day threats, and lack of explainability for security operations center (SOC) analysts.

3. Solution Architecture

Multi-layer ML pipeline:

- Layer 1 (Baseline Profiler): Isolation Forest + One-Class SVM trained on normal patterns.
- Layer 2 (Anomaly Detector): XGBoost + LSTM Autoencoder ensemble producing risk scores (0-100).
- Layer 3 (Attack Classifier): XGBoost multi-class classifier with SMOTE oversampling (7 attack types).
- Layer 4 (Explainability): SHAP TreeExplainer generating natural language explanations.

4. Evaluation Results

Accuracy: 93.6%
Precision: 75.2%
Recall: 97.7%
F1 Score: 85.0%
AUC-ROC: 0.9878

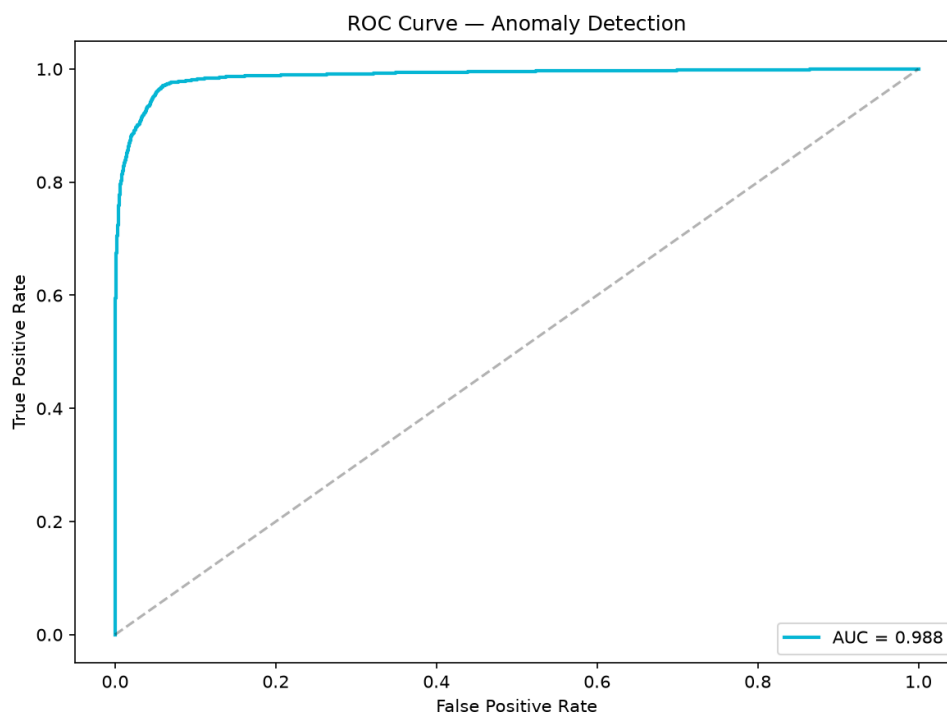


Figure 1: ROC Curve

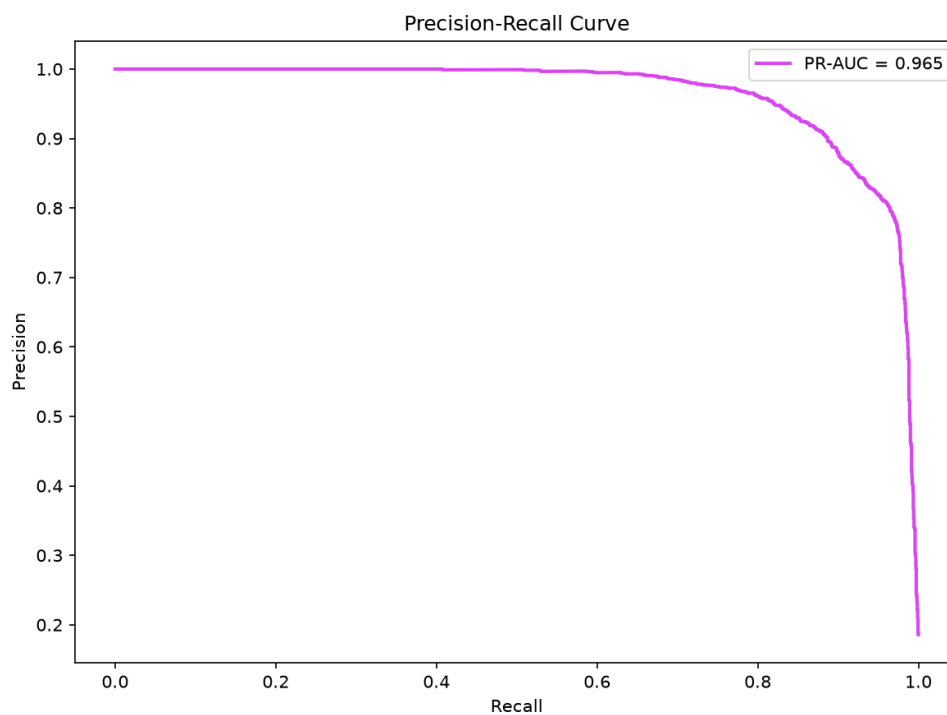


Figure 2: Precision-Recall Curve

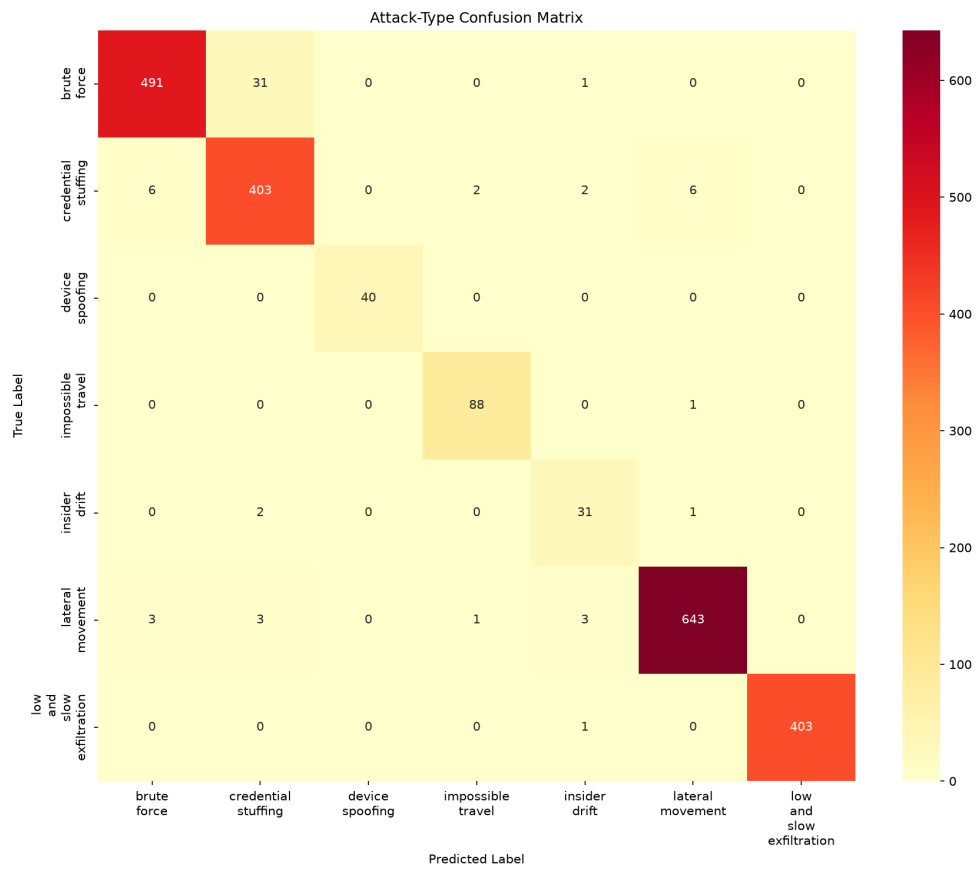


Figure 3: Multi-Class Confusion Matrix

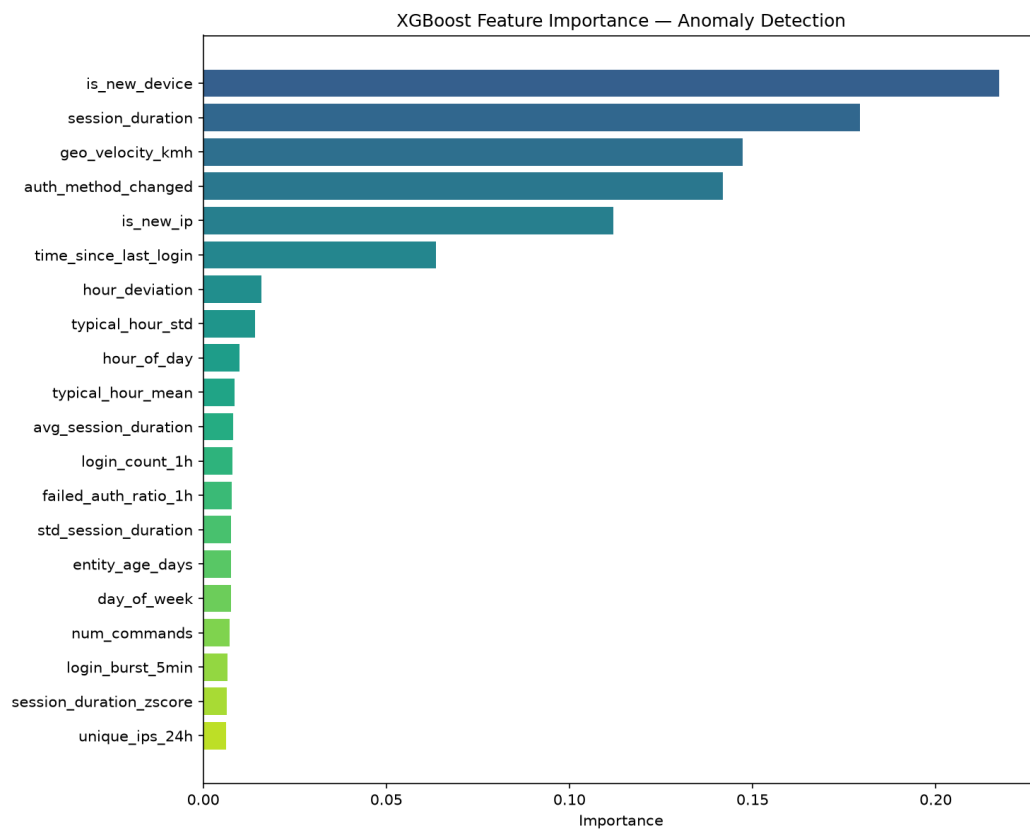


Figure 4: Feature Importance

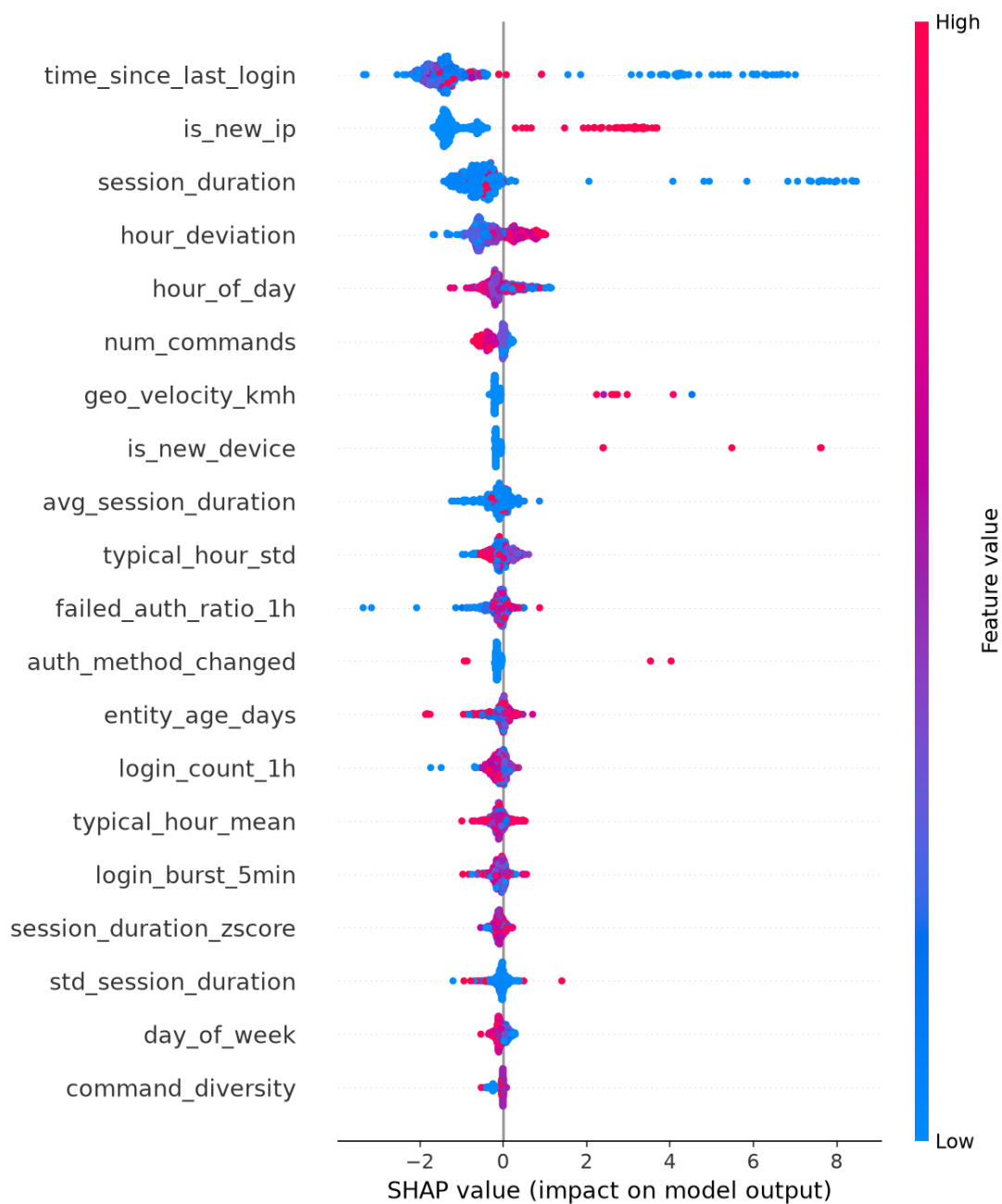


Figure 5: SHAP Global Summary Plot

5. Dashboard & Deployment Architecture

The system includes a dark/light mode analyst dashboard deployed at <https://cybershield-ai-1-5gyk.onrender.com/>.

Dashboard Functionality

- Multi-page view routing: Overview, Analytics, Alerts, Model Performance
- Global Spotlight Search (Ctrl+K / Cmd+K) across all alerts, entities, and IPs
- Light and Dark theme mode toggle with persistent preference
- 6 animated KPI counter cards with ease-out cubic transitions
- Filterable alert queue table with internal vertical scroll, pagination, and CSV Export
- Sliding AI detail panel with per-alert SHAP explanations
- Live clock, Fullscreen mode toggle, Keyboard shortcuts (1-4 for page switching)

Deployment Details

Cloud Host:	Render (Cloud PaaS)
Deployed URL:	https://cybershield-ai-1-5gyk.onrender.com/
Repository:	https://github.com/Prasanth4689/CyberShield-AI
WSGI Server:	Gunicorn
Build:	Python 3.11 with no-cache headers